

- 7 (a) (i) $-5.17 + 3.50 \text{ eV} = -1.67 \text{ eV} \Rightarrow$ Highest energy level is $n = 3$
 number of spectral line $= {}^3C_2 = 3$ [A1]

(ii) $hf = E_2 - E_1$
 $(6.63 \times 10^{-34})f = (-3.07 - (-5.17))(1.60 \times 10^{-19})$ [C1]
 $f = 5.07 \times 10^{14} \text{ Hz}$ [A1]

- (iii) 1. from ground state to $n = 2$: cannot occur [B1]
 2. from ground state to $n = 3$: can occur [B1]

The energy of the photon must be exactly equal to the energy difference between two energy levels before it can be absorbed. [B1]

- (iv) There are discrete energy levels within the nucleus. [B1]

The energy transitions are of large energies, since the radiation is in the gamma range. [B1]

- (b) (i) When high speed electrons emitted from the decay of Yttrium collides with the lead nucleus/ are deflected by lead nucleus [B1],

it undergoes large deceleration/slows down abruptly. This produces X-ray photons [B1].

The X-ray photons produced are of a range of energies as the electrons decelerates to different extent. [B1]

(ii) $E = \frac{hc}{\lambda}$
 $(100 \times 10^3)(1.60 \times 10^{-19}) = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{\lambda}$ [C1]

$$\lambda = 1.243 \times 10^{-11} \text{ m}$$

$$p = \frac{h}{\lambda}$$

$$= \frac{(6.63 \times 10^{-34})}{(1.243 \times 10^{-11})}$$
 [C1]
$$= 5.3 \times 10^{-23} \text{ kg m s}^{-1}$$
 [A1]

(iii) $\text{Force} = \frac{(1.6 \times 10^5)(5.3 \times 10^{-23} - 0)}{1}$ [C1]
 $= 8.5 \times 10^{-18} \text{ N}$ [A1]

- (c) Photoelectric experiment. [B1]

Discussion of the evidence:

- 1) The existence of a threshold frequency below which no photoelectrons are emitted proves that electromagnetic radiation (EM) consists of discrete quanta of energy given by hf . [B1]

- 2) The instantaneous emission of photoelectrons when all the photon energy is transferred immediately to the electron in a single collision gives evidence to particulate nature of EM. [B1]
- 3) The maximum kinetic energy of the photoelectrons being dependent only on frequency of radiation f , which relates to the discrete energy of photon, and independent on the intensity of radiation also give evidence for the particulate nature of EM. [B1]